Question 01 : A junior developer wrote this code to build a comma-separated list of 5,000 product IDs:

```
string productList = "";
for (int i = 1; i <= 5000; i++)
{
    productList += "PROD-" + i + ",";
}
```

Tasks:

(a) Explain why this code is inefficient. Reference what happens in memory.

(b) Rewrite this code using StringBuilder to be more efficient.

(c) Add timing code (using Stopwatch) to both versions and report the time difference.

Question 02: Ticket Pricing System

Write a program for a cinema ticket pricing system with these rules:

Condition	Price
Age < 5	Free
Age 5 - 12	30 LE
Age 13 - 59	50 LE
Age 60+	25 LE
Weekend (Fri/Sat)	Add 10 EGP to any non-free ticket
Student with valid ID	20% discount (applied after weekend surcharge)

Route
Training center & innovation space

Tasks:

(a) Implement using if-else if-else statements

(b) The program should ask for: age, day of week (1-7, where 6=Fri, 7=Sat), and whether they have a student ID (yes/no)

(c) Display the final price with a breakdown of how it was calculated

Question 03: Convert the following if-else chain to:

(a) A traditional switch statement

(b) A switch expression

```
string fileExtension = ".pdf";
string fileType;

if (fileExtension == ".pdf")
    fileType = "PDF Document";
else if (fileExtension == ".docx" || fileExtension == ".doc")
    fileType = "Word Document";
else if (fileExtension == ".xlsx" || fileExtension == ".xls")
    fileType = "Excel Spreadsheet";
else if (fileExtension == ".jpg" || fileExtension == ".png" || fileExtension == ".gif")
    fileType = "Image File";
else
    fileType = "Unknown File Type";
```

Route
Teaching center & innovation space

Question 04 : Ternary Operator

Rewrite the following using **only ternary operators** (no if statements):

```
int temperature = 35;
string weatherAdvice;

if (temperature < 0)
   weatherAdvice = "Freezing! Stay indoors.";
else if (temperature < 15)
   weatherAdvice = "Cold. Wear a jacket.";
else if (temperature < 25)
   weatherAdvice = "Pleasant weather.";
else if (temperature < 35)
   weatherAdvice = "Warm. Stay hydrated.";
else
   weatherAdvice = "Hot! Avoid sun exposure.";
```

Route
Training center & innovation space

Then answer: Is the ternary version more readable? When would you choose one over the other?

Question 05 : Input Validation with Loops

Create a **password validation** program with these requirements:

Password Rules:

- Minimum 8 characters
- At least one uppercase letter
- At least one digit
- No spaces allowed

Program Behavior:

- Use a do-while loop to keep asking until a valid password is entered
- After each invalid attempt, tell the user **which specific rules** they violated
- Limit attempts to 5. After 5 failed attempts, display "Account locked" and exit
- On success, display "Password accepted!"

Hint: Use foreach to iterate through characters and check conditions.

Route
Training center & innovation space

Question 06 : Array Processing

Given an array of exam scores:

```
int[] scores = { 85, 42, 91, 67, 55, 78, 39, 88, 72, 95, 60, 48 };
```

Using loops (your choice of for, foreach, while), write code to:

(a) Find and display all **failing scores** (below 50)

(b) Find the **first score above 90** and stop searching immediately

(c) Calculate the **class average**, excluding any scores below 40 (considered absent)

Route
Training center & innovation space

(d) Count how many students scored in each grade range:

- A: 90-100
- B: 80-89
- C: 70-79
- D: 60-69
- F: Below 60